Room plan: tower cities (how many towers can you see?)

The plan for Wednesday, November 11, 2026 (proposed; James has not chosen it yet). Week 5, Tower cities: packets for K–1, 2–3 and 4–5, and the facilitator guide, whose launch this plan moves to the floor. Unpiloted.

Why this follows November 4

On October 29 the children counted the orders of 3 and 4 cards and built orders from swaps; on November 4 (planned) a shuffle machine moves a whole order at once. A row of towers of different heights is an order of the heights too: the same object, read a new way, by how many towers you see from an end. The opener's animal line is such an order.

For adults: a row of towers 1 to n tall is a permutation, and the towers seen from an end are its records. The tallest is seen from both ends and every other seen tower stands on its own side of it, so the two counts satisfy $L + R \leq n + 1$, and 1 and 1 never happens. A city is a Latin square with these counts as edge clues: a clue set can fit one city, several or none; every 3-by-3 city's twelve numbers add to 22; the fewest clues that fix a city are 2 for 3 by 3 and 3 for 4 by 4 (by computer); and 204 of the 576 4-by-4 cities share all sixteen numbers with another, so no puzzle singles them out (4–5 Problems 4, 5 and 9).

Prep checklist for Wednesday afternoon

Print single-sided, US Letter, at 100% ("Actual size"), so cubes stand on the 1-inch squares (2–3 pages 2–7, 4–5 page 2). Counts assume about 4 children at K–1, 4 to 6 at 2–3 and 3 at 4–5.

- K–1 packet, 7 pages: 5 copies (one for the adult who reads aloud)
- Grades 2–3 packet, 9 pages: 7 copies, plus 4 extra of page 7 (extra grids for Problem 5)
- Grades 4–5 packet, 8 pages: 4 copies, plus 6 extra of page 2 (the only 1-inch 4-by-4 grid; puzzles are copied onto it to build)
- Facilitator guide, 10 pages, and this room plan: one each per adult (3)
- Opener sheet, 2 pages, cut in half: page 1 (main, two per page) 5 copies, page 2 (easier, harder) 3 copies
- Signs on paper: MOUSE, CAT, DOG, HORSE, GIRAFFE, and LEFT END
- Snap cubes (3/4 inch), the lighter route below: about 155
- 7 file folders as screens (2 per table, 1 for the floor); the big whiteboard laid flat as a floor base, so towers stand on carpet
- Small whiteboards, pencils, plain paper (laid lengthwise off a row's end, it marks "about a foot"), a big timer
- Ten-minute test the evening before: the guide's three tests, plus one untried: the launch towers on the floor base as 2, 1, 4, 5, 3 show 3 from the left and 2 from the right, chin on the floor a foot away

The cubes: two routes. The guide pre-builds a set for every child (its "main job"). The lighter route keeps the mathematics: one set per pair, children build their own towers inside Problem 1 or 2 (as the problems say), and one colour per height at 2–3 and 4–5 (1 yellow, 2 blue, 3 red, 4 green; 5 purple, launch only), so a repeated colour in a line shows a mistake. The 2–3 pairs play Problem 5 against each other; 4–5 keeps two sets because the guide's referee builds second cities.

Table	Guide: own sets, pre-built	Lighter: shared sets, loose cubes
K–1 (4)	24 + 6 spare, 12 towers	24, a bag of 6 per child, any colours
2–3 (4)	72 + 10 spare, 36 towers	36: per pair 3 yellow, 6 blue, 9 red
4–5 (3)	120 + 10 spare, 48 towers	80: two sets of 4 yellow, 8 blue, 12 red, 16 green
Launch	borrows a K–1 set	15: towers 1 to 5, built; stays on the floor
Total	**242 cubes, 96 towers built**	**155: 15 yellow, 30 blue, 45 red, 36 green, 5 purple, 24 any**

Table	Guide: own sets, pre-built	Lighter: shared sets, loose cubes
Prep (untimed estimate)	45 to 60 minutes	20 to 25 minutes (30 from a mixed bin)

Add 18 cubes per extra 2–3 pair and 6 per extra K–1 child. If the bin is short, give 4–5 one set (115 in all) and the referee checks in pencil on a page-2 grid. November 4 uses about 75 cubes from the school's bin, whose size nobody knows: count it by colour that day. Building costs children about 2 minutes at 2–3 and 3 to 4 at 4–5.

Before the children arrive

- Opener half-sheets and pencils on every table. The floor spot: the base, a sheet of paper lengthwise off each end, LEFT END on the floor at the room's left as the room sees it, the animal signs, the five launch towers and one folder.
- At each table: the cubes, that band's pages face down, whiteboards, two folders, plain paper, and a foot of clear table beyond every row (guide).
- Early-finisher cards on index cards. K–1: "With your partner's towers, build a row of 1, 2, 3, 4 that shows 3 from the left and 2 from the right." (1243, 1342 or 2341.) 2–3: "Build a city with three or more 3s around its edge, or say why you can't." (Every 3-by-3 city has exactly two.) 4–5: "Make a 5-tower. How many rows of 1 to 5 show 3 from both ends?" (6: the 5 in the middle.)

Opener: line up for the photo (8 minutes)

On the tables as children arrive, then acted out on the floor so its answer becomes the launch's row. An adult reads clues aloud at any table. The sheet pictures the heights: Mouse 1 cube, Cat 2, Dog 3, Horse 4, Giraffe 5.

The puzzle (most of the room). Five animals stand in a line for a photo. Write each animal's name in its place in the line.

1. Horse stands in the middle.
2. Dog stands at one end of the line.
3. Mouse stands between two taller animals.
4. Mouse stands right next to Cat.
5. The animal at the left end is shorter than the animal at the right end.

Answer: Cat, Mouse, Horse, Giraffe, Dog (heights 2, 1, 4, 5, 3). It is the only answer, and every clue is needed (checked by computer, photo-line_check.py).

Hint ladder (give one step, then walk away):

1. "Which clue tells you exactly where one animal stands? Write that one first." (Horse, the middle.)
2. "Every animal is taller than Mouse. So where can Mouse stand?" (Second or fourth.)
3. "Try Mouse second: where do Cat and Dog go? Now try Mouse fourth. Which passes clue 5?" (Mouse fourth puts Dog left and Cat right: clue 5 fails.)

Easier. Mouse, Cat and Dog: the tallest animal stands at the right end; Mouse stands between two taller animals. *Answer:* Cat, Mouse, Dog, the launch's row.

Harder. The same five, and no clue names a place: every animal taller than Cat stands somewhere to the right of Cat; Cat stands right next to a shorter animal; Horse stands right next to Mouse; the two tallest animals stand side by side. *Answer:* the same line; every clue is needed.

Early finishers. "Cross out clue 5. How many answers now? Write a new clue that makes it exactly one again." (Two: the line and the same line turned round, since clue 5 alone tells left from right. "Giraffe stands somewhere to the right of Horse" fixes it, and so do others.)

On the floor (minute 6). Five solvers take the signs and towers and stand in the answer order facing the room, Cat by LEFT END. The room checks each clue, thumbs up or down. The bridge question: "A photographer lies down at the left end, chin on the floor. How many animals are in the photo?" Take guesses; the five set their towers on the base in order, and one child looks from the end of the paper: 3 (Mouse hides behind Cat, Dog behind Giraffe). From the right end: 2.

The hour

Time	Everyone	Jobs
0:00–0:08	**Opener: line up for the photo.** On the tables as children arrive; at minute 6, five solvers stand in the answer order and a child checks the photo with a chin on the floor.	Solvers check sheets; adults give one hint step at a time.
0:08–0:11	**Launch on the floor**, the guide's launch with the opener's towers. Take Horse and Giraffe away: Cat, Mouse, Dog is the guide's 2, 1, 3 (2 from the left, 1 from the right). "Change the row so that you see 3 from this end." Then the folder: hide a row, say its two numbers, pointing to each end. Practise the reset signal: "Hands off the towers", hands on heads.	One child moves towers, another checks chin down; everyone else predicts out loud.
0:11–0:19	**Tables, round 1.** Pairs build only what each problem needs. K–1: Problems 1 and 2 (six rows; two frames stay empty). 2–3: Problem 1. 4–5: Problem 1 (eight cards). At the timer, one pair shows one row to its table.	**Builder** stands towers; **Looker**, chin down, says and writes the numbers. Swap each card.
0:19–0:27	**Round 2.** K–1: Problem 3 (two of the six cards get an X). 2–3: Problem 2 (nine towers in the colour code, two cities). 4–5: Problem 2, then Problem 3's first two puzzles on page-2 grids.	2–3: **Builder** and **Checker**, swap each city. 4–5: a pair and a rotating **Referee**.
0:27–0:33	**Back to the floor: hidden row of four.** The organizer hides Dog, Mouse, Horse, Cat (3, 1, 4, 2) behind a folder and says "2 and 2", pointing. Each table sends a builder with towers 1 to 4 (K–1 and 2–3 snap a 1 onto a 3). Lift it: different rows, all right (six rows show 2 and 2).	Builders build; their tables check chin down.
0:33–0:41	**Round 3.** K–1: Problem 4, the hidden-row game (this table's main activity). 2–3: Problem 3 (puzzle 3 has no city), then Problem 4. 4–5: Problem 3 on, then Problem 4 (two of its four cities share all sixteen numbers).	K–1: **Hider** and **Builder**; 2–3: **Solver** and **Checker**; swap each puzzle.
0:41–0:48	**Round 4.** K–1: Problem 4 again, or Problems 5–7 with pooled cubes. 2–3: Problem 5, pair against pair. 4–5: Problem 5 (hidden city), and Problem 9 if twins turn up.	Swap hiding and solving after each puzzle.
0:48–0:55	**Floor finale: two photographers**, chin down at each end of the five animal towers. "Make each photographer see just 1." (Impossible: Giraffe is in both photos; the guide's sharing question.) "Now both see 3." (Giraffe in the middle, as Mouse, Horse, Giraffe, Dog, Cat.) "Both see 4?" (No: that takes seven animals.)	A tower mover per table; new photographers each try.
0:55–1:00	Towers back in their cups as they are; pages collected.	Each pair counts its towers by height.

If it goes sideways

- A table finishes early: its early-finisher card, then later problems (K–1 5–7, 2–3 6–7, 4–5 6–9). A pair is stuck: "Chin down. Which tops can you see?", nothing more.
- Counts disagree: eye at the end of the paper, a foot away; slide a city's middle row out to look (guide). A table runs out of attention before 0:48: its fallback in the guide (K–1 the adult hides a row; 2–3 a hidden row of four; 4–5 fewest questions).
- 4–5 Problem 5: a second city with all the same numbers is not a mistake but twins (Problem 9). The room gets loud: reset signal, then the next floor activity.
- Returning children: the use log shows nobody doing tower cities, at the circle or at home (Jay and Rose did shuffle machines at home). If a child knows the app's tower puzzles, make them a photographer at 0:48 and a checker who may say "try again" but not why.

How it went

(Add a dated line after the session.)